KLAi

Exploring and visualising data with Python: from a new file to a finding

How do I explore an unfamiliar dataset in one sitting and find what is worth reporting?

AI and Data Science Professional | Session 11 of 24 | Module C: Analyse and explain

KLAi Professional Learning | Jotham Martin Wambi

Today in one view: what you will leave with

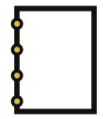
By the end of today you can



Run a repeatable exploratory data analysis (EDA) routine on a new dataset

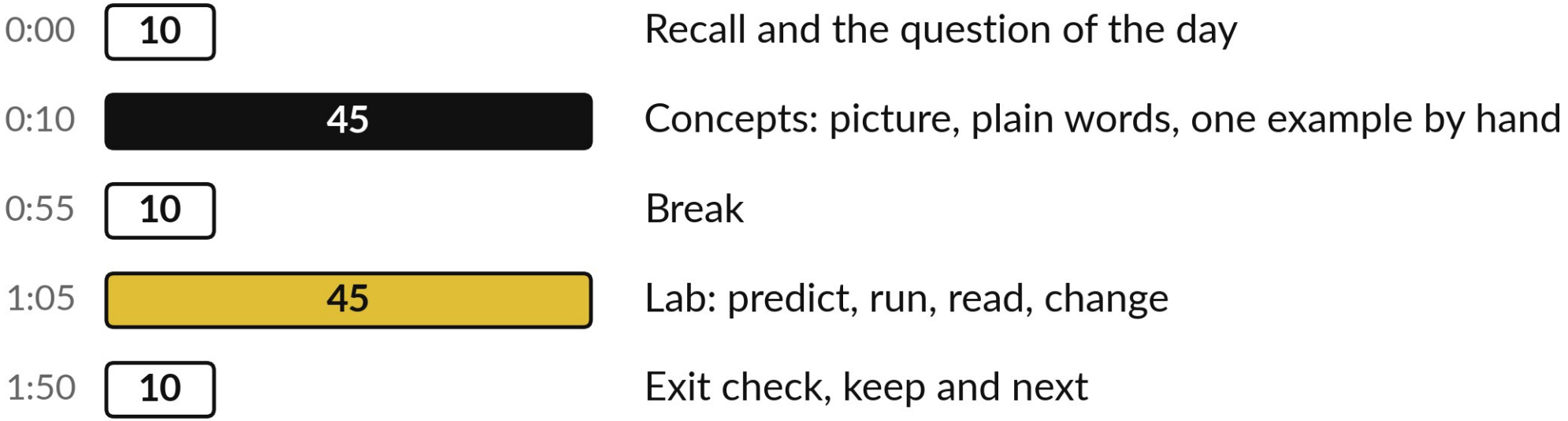


Draw histograms, box plots, scatter plots, bar charts and line charts with matplotlib and seaborn



Tell a count from a rate and spot seasonality, then write a finding with a number, a comparison and a limit

Your two hours: concepts first, then a Python lab in Colab



Pictures and ideas first. Then you open the notebook and draw your own charts.

Three questions from last time

1. In SQL, what is the difference between WHERE and HAVING?

2. Why did you check each SQL result against pandas?

3. Which SQL clause turns many order lines into one number per branch?



Answer aloud first. Then we check together.

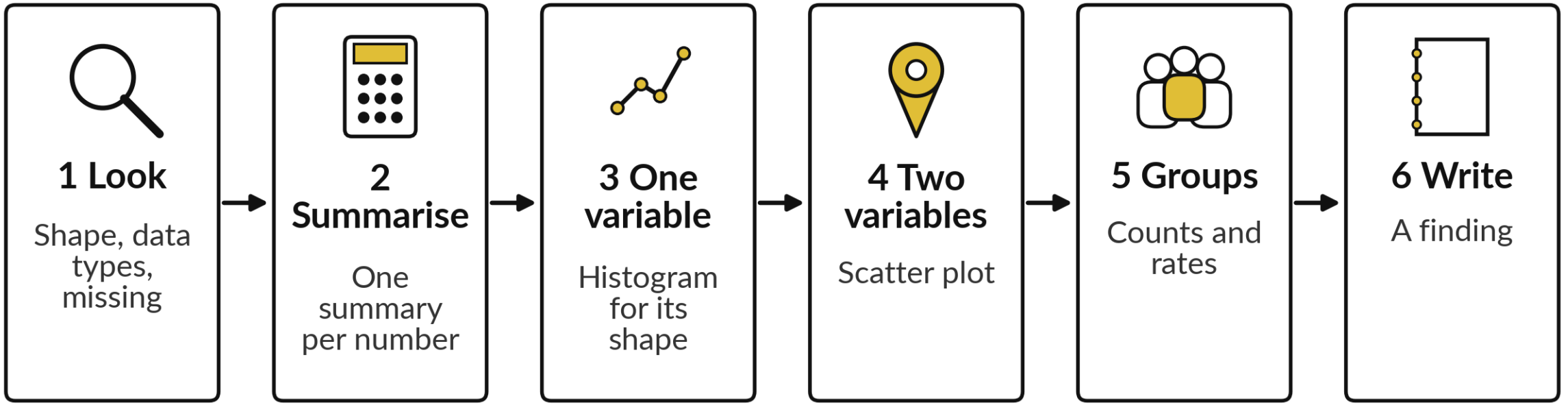
Answers to the three questions from Session 10

Question	Short answer
In SQL, what is the difference between WHERE and HAVING?	WHERE keeps rows before the grouping. HAVING keeps groups after the aggregation, such as a count or a sum.
Why did you check each SQL result against pandas?	Two different routes to the same number is the best evidence that the number is right.
Which SQL clause turns many order lines into one number per branch?	GROUP BY, together with an aggregate function such as COUNT or SUM.

Words we use today

	Word	What it means
	**Exploratory data analysis (EDA)**	A repeatable routine of looks, summaries and charts that you run on a new dataset before you model it or report on it.
	**Figure and Axes (matplotlib)**	The Figure is the whole page. The Axes is the plot area on that page where the data are drawn and labelled.
	**Count and rate**	A count is how many. A rate divides the count by what could have happened, such as returns per 100 lines sold.
	**Seasonality (with trend and noise)**	A pattern that comes back at the same time every year, such as a December peak.
	**Finding**	One sentence that states a number, a comparison and a limit.

Exploratory data analysis (EDA) is a six-step routine you repeat on every new file



Exploratory data analysis (EDA) is what you do in Excel when you scroll, filter and pivot a new sheet. Python lets you write the routine down so it runs the same way every time.

Step 1, the first look: read the file, then count the records and the missing values

```
import pandas as pd
orders = pd.read_csv('orders.csv',
                     parse_dates=['order_date'])
print(orders.shape)
orders.isna().sum()
```

```
(30229, 7)
order_id            0
order_date          0
branch_id           0
customer_id      9171
payment_method      0
promo_week          0
order_value_ugx     0
```

In plain words

read_csv loads the file into a DataFrame. parse_dates makes order_date a real date.

shape: 30,229 records and 7 variables.

isna().sum() counts the missing values in each variable.

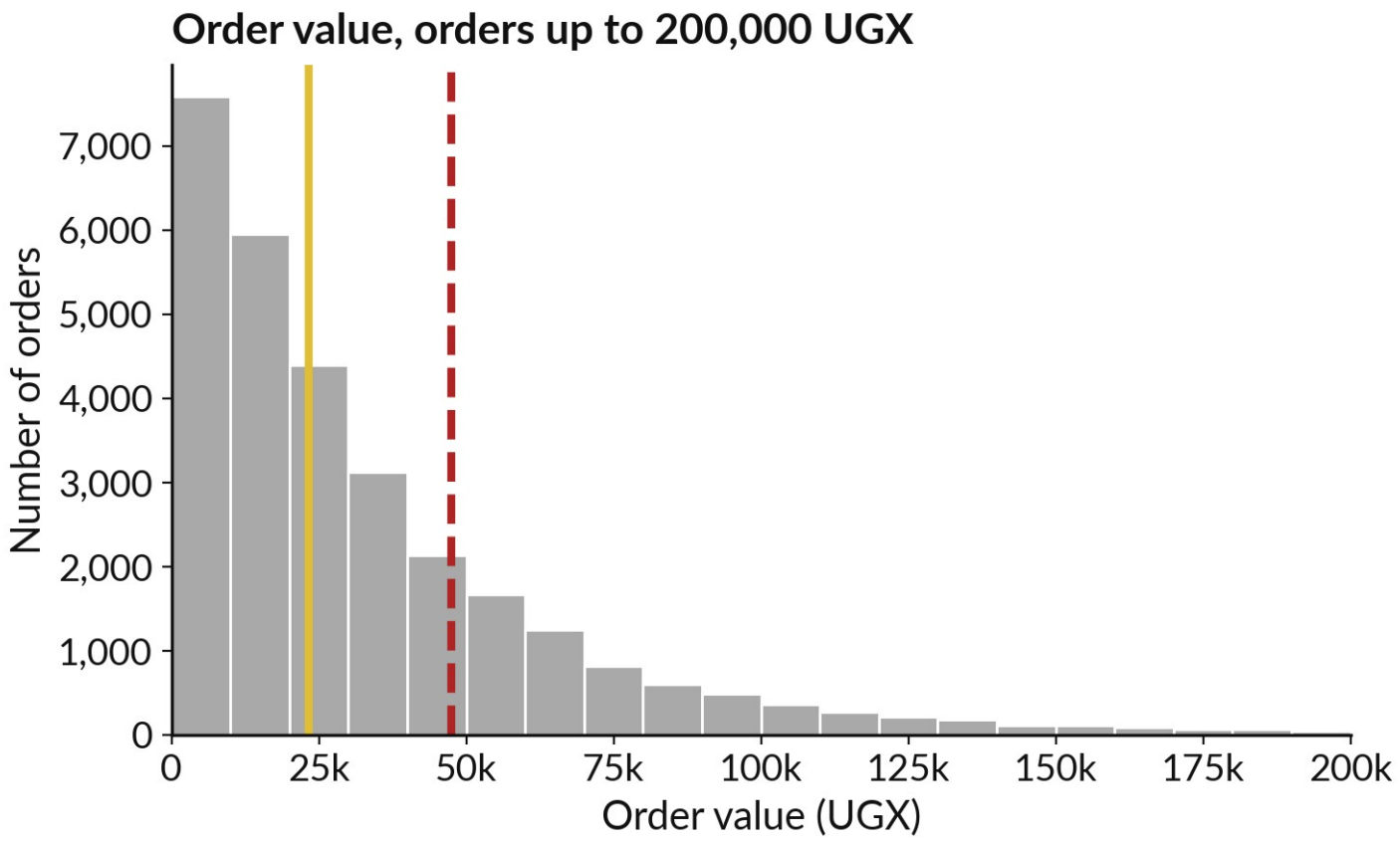
customer_id has 9,171 missing: walk-in shoppers.

Step 2, summarise: the mean order value is 2.0 times the median, so the distribution is right-skewed

Statistic	Order value (UGX)	What it tells you
count	30,229	How many orders have a value.
mean	47,491	The arithmetic mean: the total divided by the count.
std	123,574	The standard deviation: the typical distance from the mean.
min	800	The smallest order.
25% (Q1)	9,940	A quarter of the orders are below this.
50% (median)	23,200	Half of the orders are below this.
75% (Q3)	47,300	Three quarters of the orders are below this.
max	3,104,000	The largest order.

Python: the describe method. Excel: AVERAGE, STDEV.S, MIN, QUARTILE, MEDIAN, MAX.

Step 3, one variable: a histogram shows the shape of the distribution



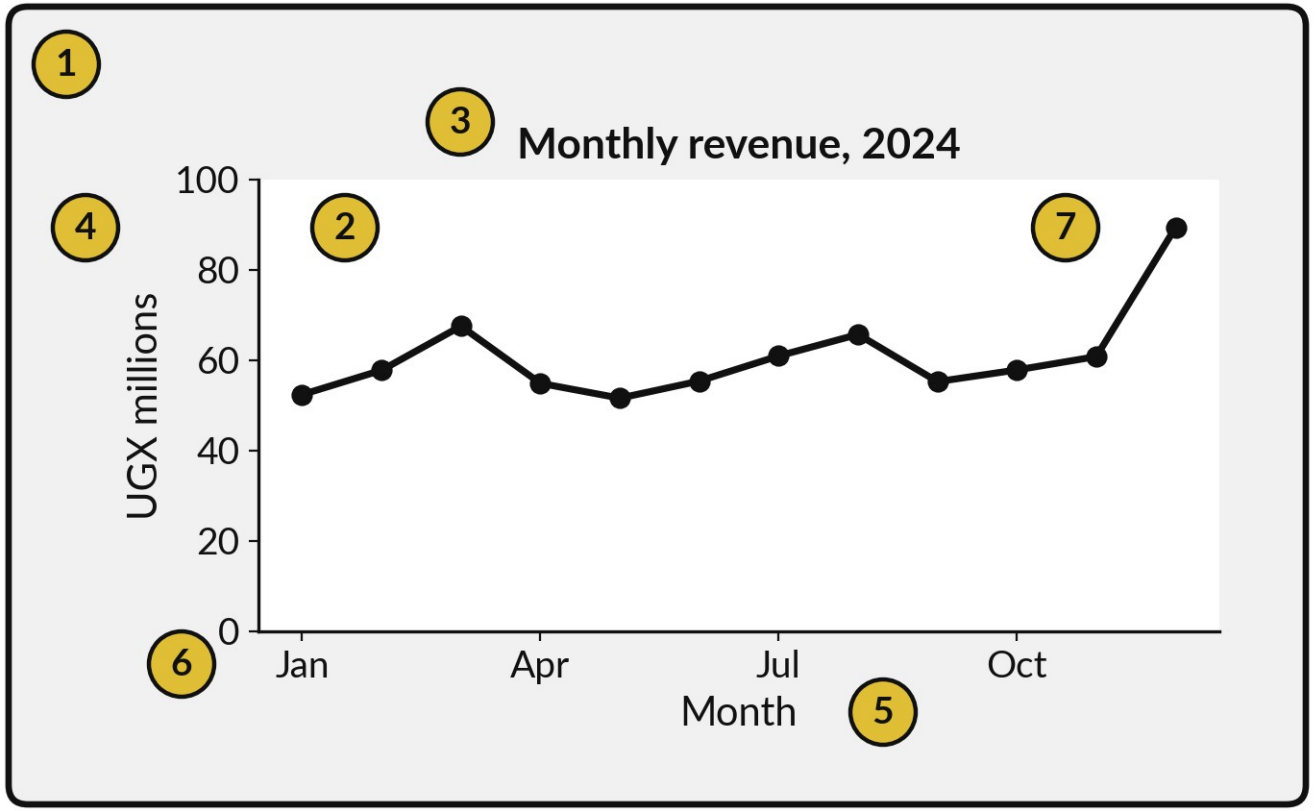
A histogram counts how many records fall in each bin (a range of values).

Tall bars on the left: most orders are small.

Long right tail: the mean 47,491 is right of the median 23,200.

720 orders above 200,000 are off the chart.

matplotlib draws on a Figure that holds an Axes



1. **Figure (fig): the page**
   `plt.subplots()`
2. **Axes (ax): the plot area**
   `same call`
3. **Title**
   `ax.set_title()`
4. **y-axis label**
   `ax.set_ylabel()`
5. **x-axis label**
   `ax.set_xlabel()`
6. **Tick labels**
   `automatic`
7. **Line: the data**
   `ax.plot()`

An axis is one scale. The Axes holds both.

Draw a line chart with matplotlib: make the Figure and Axes, draw, label

```
import matplotlib.pyplot as plt
value = orders['order_value_ugx']
by_month = orders['order_date'].dt.to_period('M')
monthly = value.groupby(by_month).sum() / 1e6
fig, ax = plt.subplots()
ax.plot(monthly.index.astype(str), monthly.values)
ax.set_title('Monthly revenue')
ax.set_ylabel('UGX millions')
print(type(fig).__name__, type(ax).__name__)
```

```
Figure Axes
```

In plain words

Total the order value for each month and divide by a million, so the axis reads in UGX millions.

plt.subplots() returns the Figure and the Axes.

ax.plot draws one line. set_title and set_ylabel add the words.

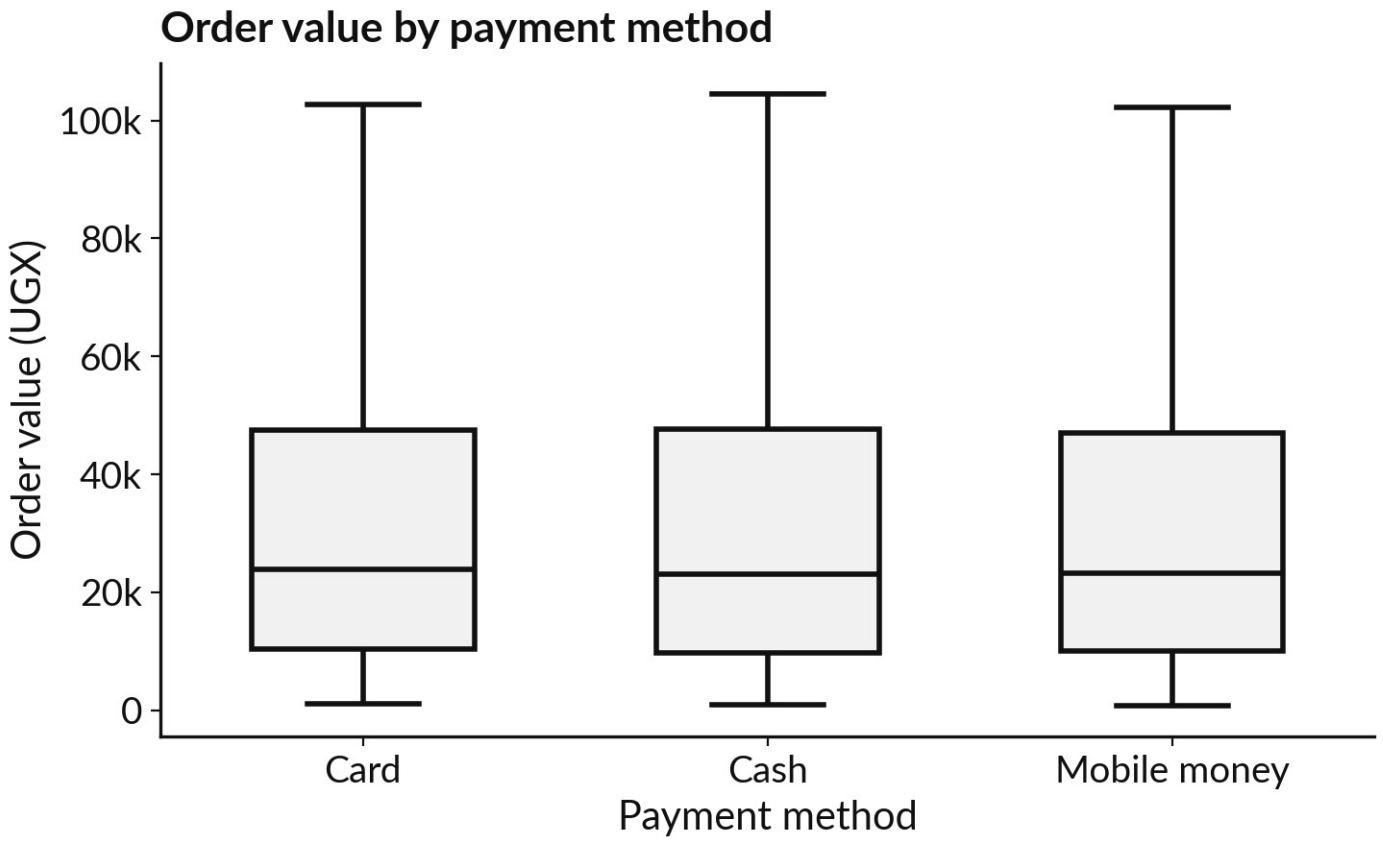
The last line checks the types: a Figure and an Axes. In Colab the chart appears under the cell.

Choose the chart from the question, then the seaborn function that draws it

	Question you ask	Chart	seaborn function	Lakeside example
	**How is one numeric variable spread?**	Histogram	sns.histplot	Order value, basket size
	**How does a numeric variable differ between groups?**	Box plot	sns.boxplot	Order value by payment method
	**Do two numeric variables move together?**	Scatter plot	sns.scatterplot	Basket size and order value
	**Which group has the larger mean or count?**	Bar chart	sns.barplot	Mean order value by branch
	**How does a number change over time?**	Line chart	sns.lineplot	Revenue by month

seaborn builds on matplotlib. You pass a DataFrame and the column names, and it returns an Axes. hue colours the marks by a categorical variable.

A box plot compares the spread of order value across payment methods



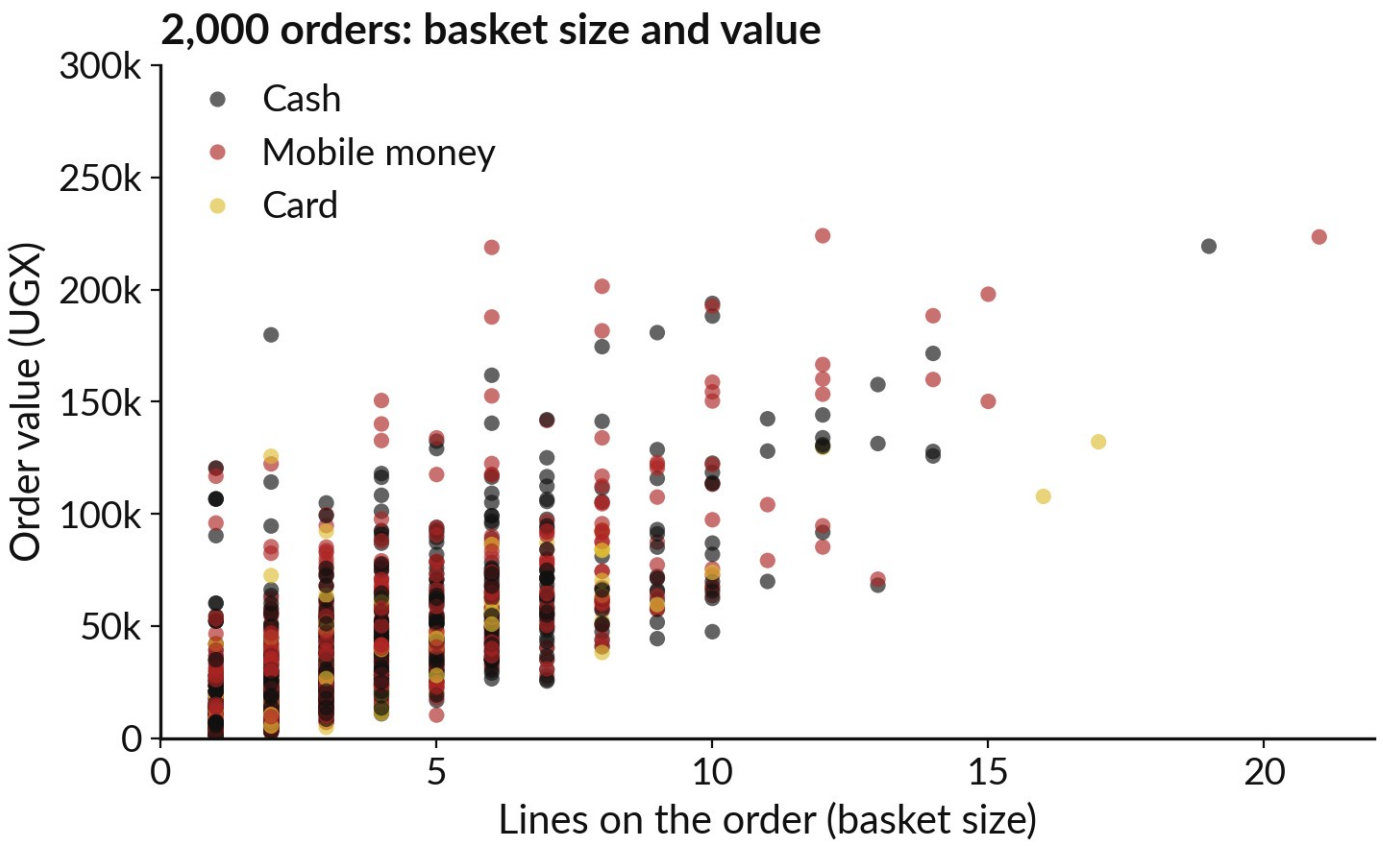
Each box holds the middle half of the orders, from Q1 to Q3. The line inside is the median.

Whiskers end at the last order within 1.5 x IQR of the box.

About 2,100 orders (6.9%) are above the fence, 103,340 UGX.

Medians within 800 UGX. Each box fence is within 1,206 of it.

Step 4, two variables: a scatter plot shows whether they move together



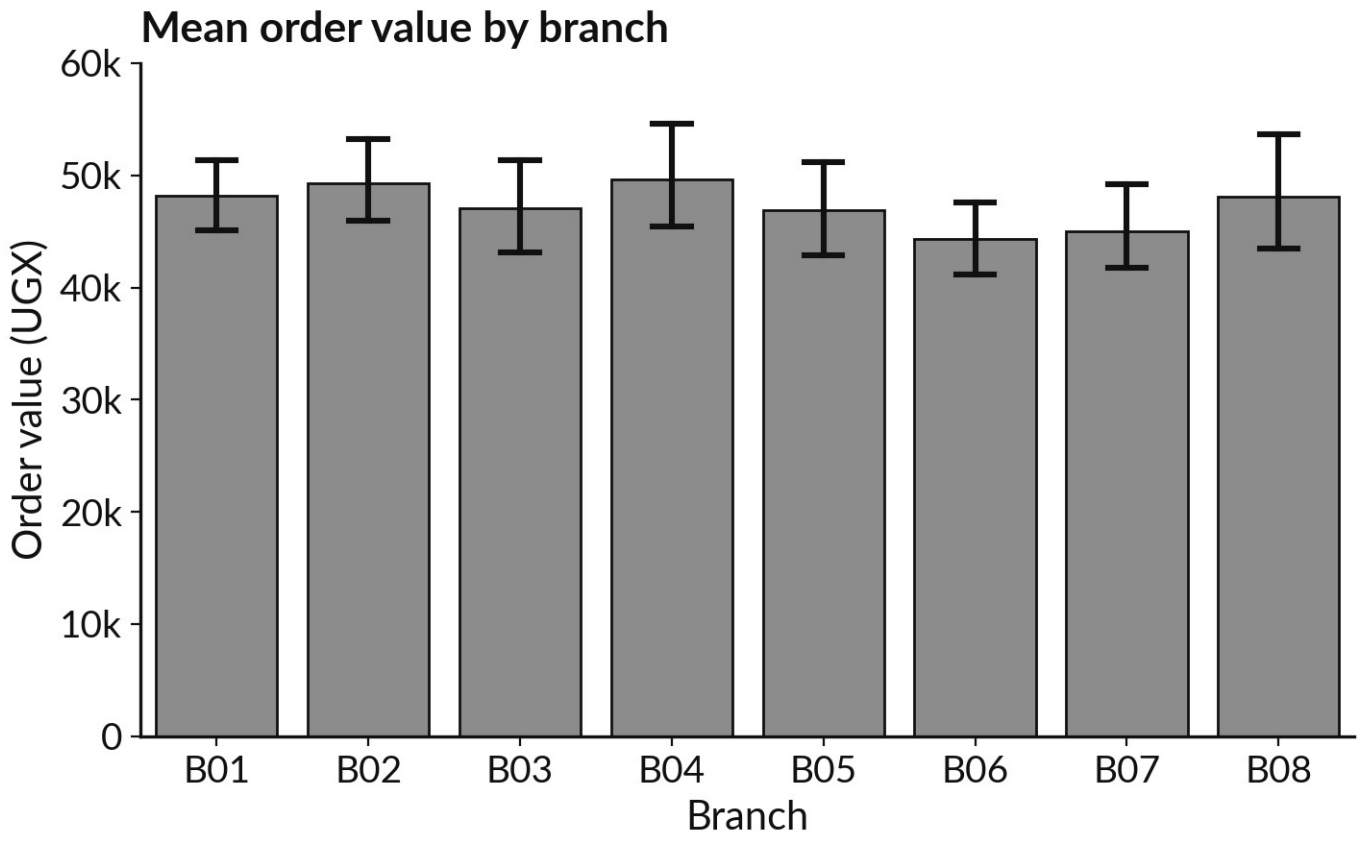
Bigger baskets cost more. The correlation is r = 0.30.

r runs from -1 to 1. About 0.3 is modest.

hue colours points by payment method. No colour stands apart.

36 points above 300,000 are off the chart.

A bar chart of means needs its error bars: the branches barely differ



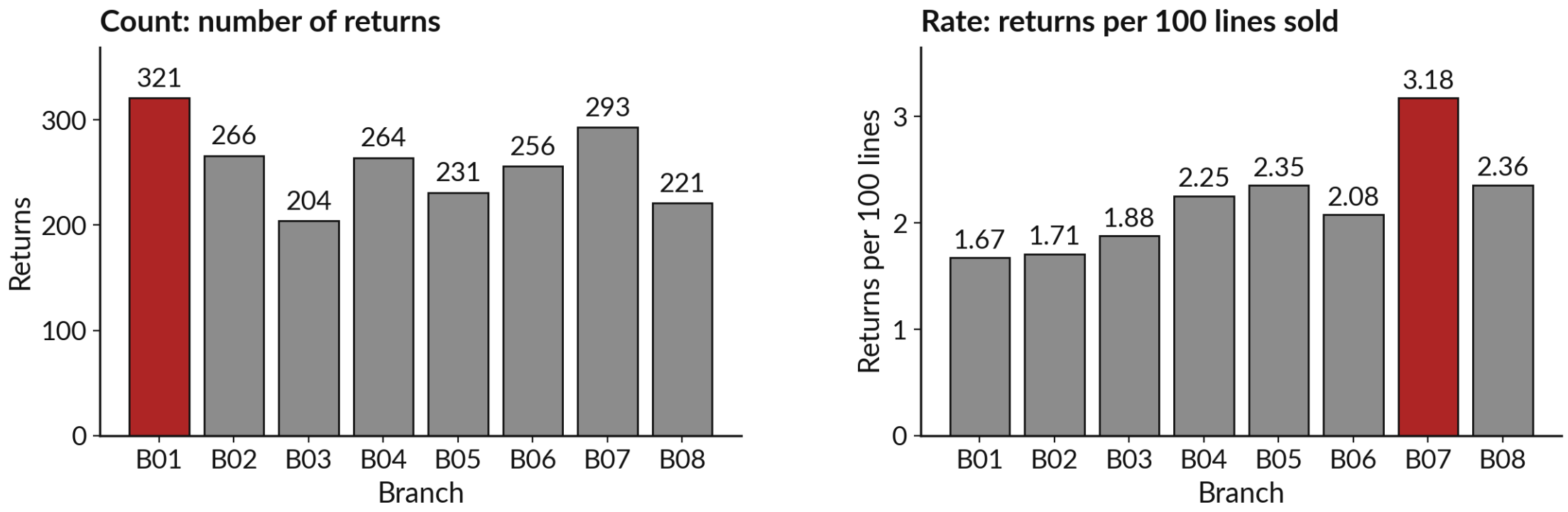
sns.barplot draws each group mean. The black line is a 95 percent confidence interval.

Highest mean B04: 49,618. Lowest B06: 44,374.

The intervals overlap: the gap may be sampling variability. This is no test.

Session 12 tests it properly.

Step 5, groups: B01 has the most returns, but B07 has the highest rate



Red marks the top bar. B01 is Kampala Central. B07 is Fort Portal.

A rate divides the count by what could have happened: returns per 100 lines sold

Rate

$$\text{rate} = \frac{\text{count of returns}}{\text{lines sold}} \times 100$$

Branch B07

$$\frac{293}{9227} \times 100 = 3.18$$

Symbol	What it means	In our data
count	How many events happened.	293 returns at B07
denominator	What could have happened: the lines sold.	9,227 lines at B07
x 100	Gives events per 100 lines.	3.18 per 100
All branches	Total returns over total lines, times 100.	2,056 / 98,066 = 2.10

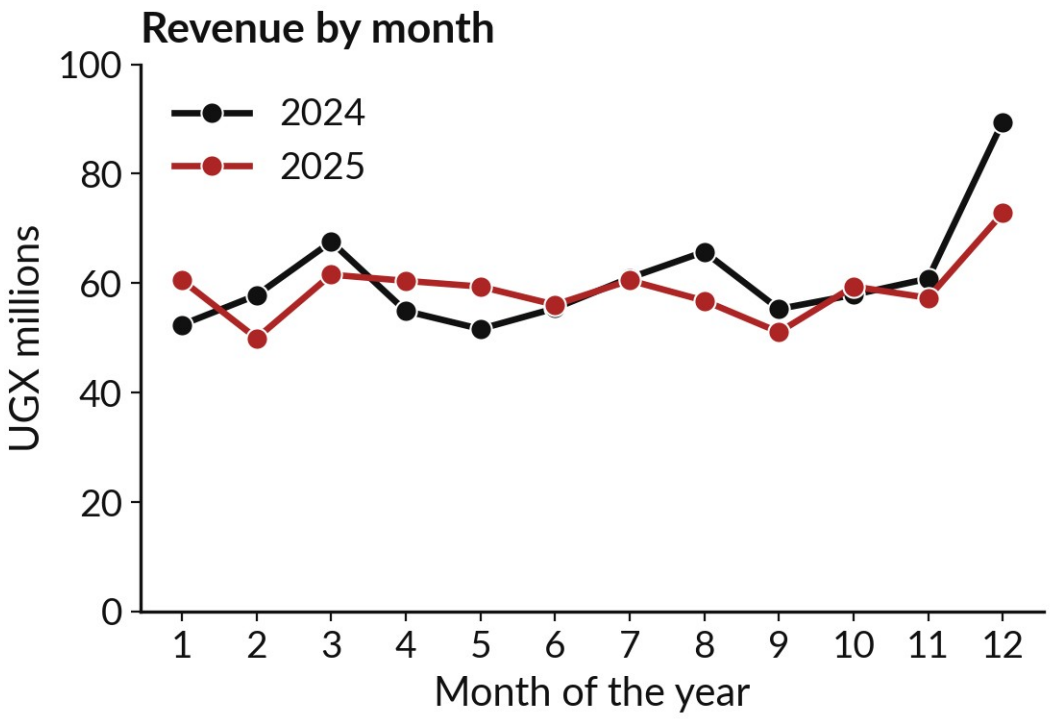
A rate is a ratio of events to opportunities. A percentage is a rate per 100.

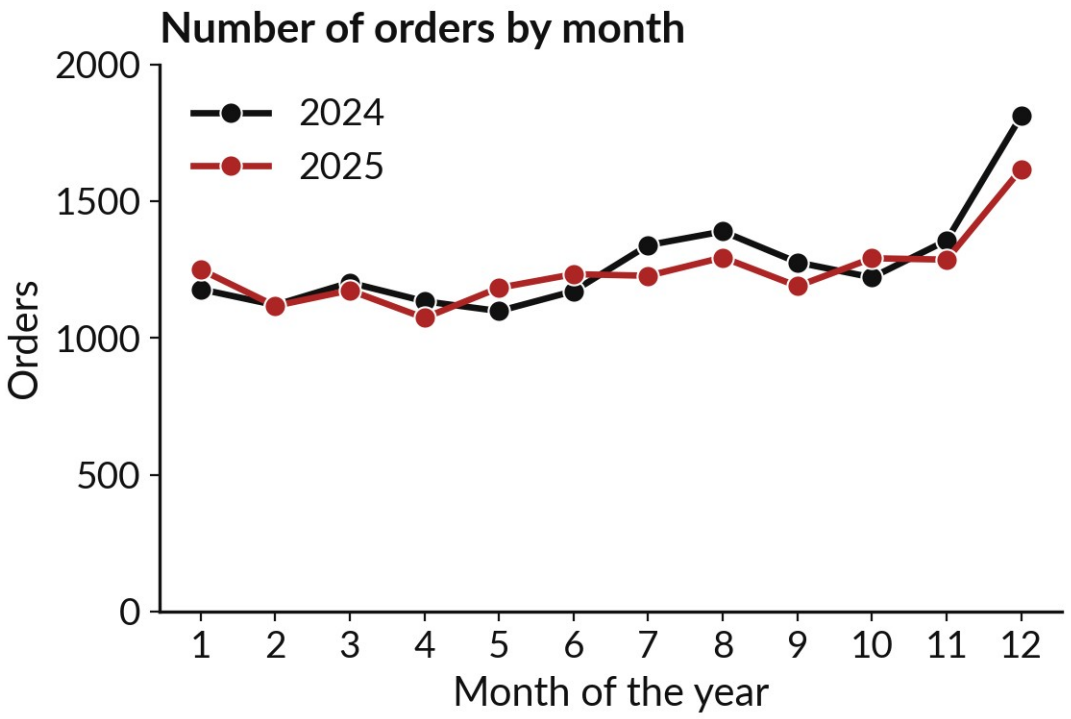
The ranking reverses: B01 is first by count and last by rate

Branch	Returns (count)	Lines sold	Rate per 100 lines	Rank by count	Rank by rate
B01 Kampala Central	321	19,169	1.67	1	8
B02 Kampala Ntinda	266	15,567	1.71	3	7
B03 Entebbe	204	10,862	1.88	8	6
B04 Jinja	264	11,718	2.25	4	4
B05 Mbale	231	9,812	2.35	6	3
B06 Mbarara	256	12,330	2.08	5	5
B07 Fort Portal	293	9,227	3.18	2	1
B08 Gulu	221	9,381	2.36	7	2

Rank 1 is the highest. Returns are counted in the returns file and lines sold in order_lines (lines, not units), both joined to the branch of the order.

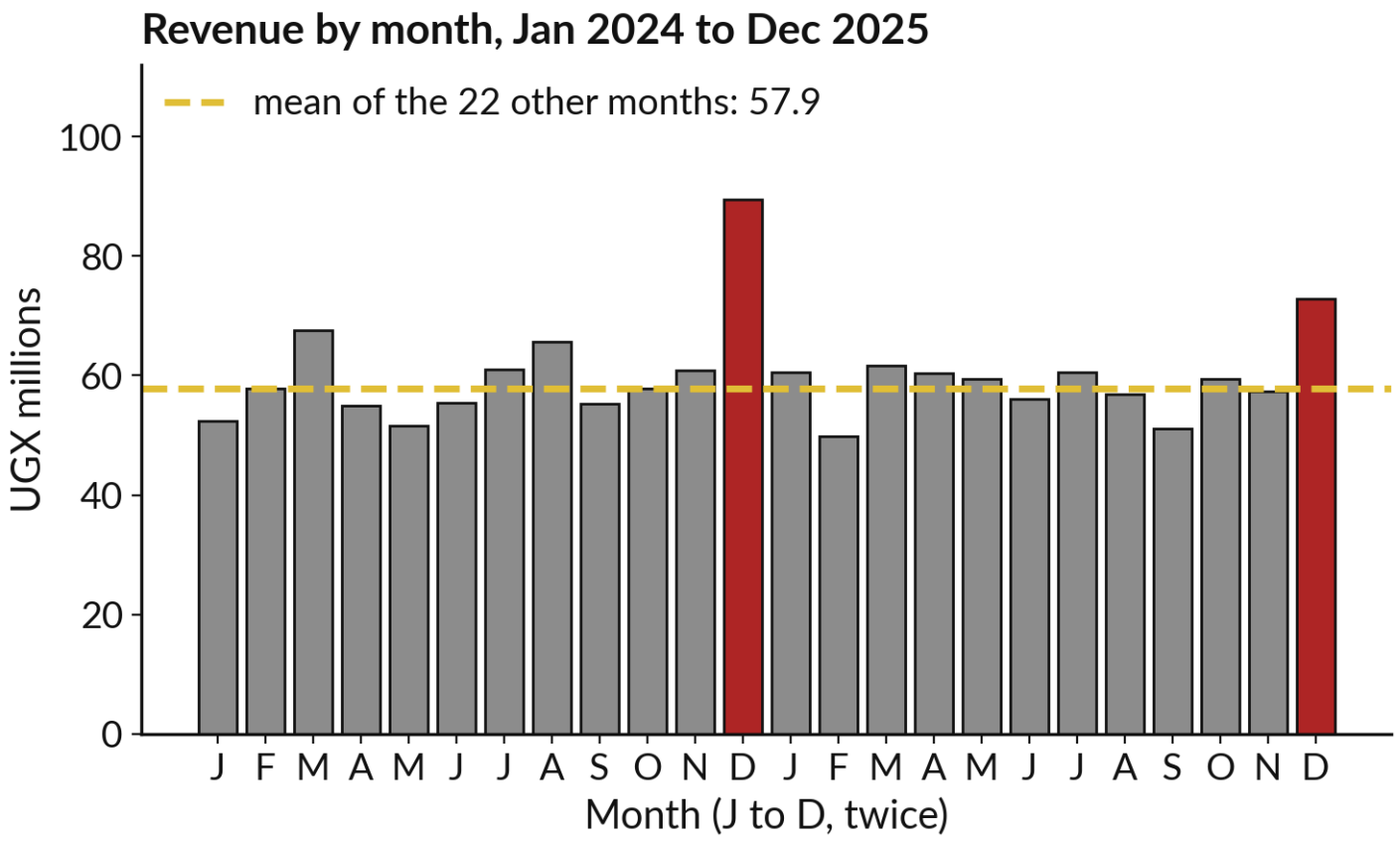
Seasonality: revenue peaked in December in both years, because there were more orders





Each line is one year over the same twelve months. Both years rise in December.

December revenue was 1.55 and 1.26 times the average of the other months



A time series has a trend, seasonality and noise. This one has a clear seasonal peak.

Trend: slow rise or fall.
Seasonality: yearly repeat.
Noise: random variation.

Mean of the 22 other months: 57.9 million.

Limit: only two Decembers.

Step 6, write: a finding states a number, a comparison and a limit

Weak

Returns are high in Kampala Central.

No number, comparison or limit.

Strong

Kampala Central has the most returns (321) but the lowest rate: 1.67 per 100 lines sold, against 2.10 overall.

Limit: a 5 percent sample, 2024 to 2025.

If you cannot write the sentence, you have not finished exploring.

Five habits make a chart honest before you send it

	Habit	Why	Example from today
	**Title that states the point**	The reader should not have to guess.	Revenue by month, with both years
	**Axis labels with units**	A number without a unit has no meaning.	UGX millions, returns per 100 lines
	**Bars start at zero**	A truncated axis makes small gaps look large.	The branch means chart starts at 0
	**Say what hidden values exist**	Limits on an axis can hide records.	720 orders above 200,000 UGX
	**Explain every colour**	Colour needs a legend or a caption.	Red marks the top bar; hue is payment method

Where you will use this at work: explore a new file in the same order every time

	Your job	The new file	What today lets you do
	**Credit analyst at a bank in Kampala, using pandas and seaborn**	A loan book of 8,000 loans.	Check missing incomes and loan sizes before any model.
	**District health officer preparing a monthly report**	Clinic visits by district and month.	Plot each district by month, spot the seasonal peak, and report visits per clinic day, or a rate per 1,000 people when you have the population.
	**Retail analyst handing charts to a manager**	Till orders and returns by branch.	Show rate and count side by side, with a finding and its limit. Export with savefig and to_excel.
	**Programme officer at an NGO**	A survey export with text and numeric answers.	Run the six steps in a notebook a colleague can rerun.

Read the recipe before the lab: merge, draw, check

```
# run once; to run again, first reload orders
import seaborn as sns
lines = pd.read_csv('order_lines.csv')
basket = lines.groupby('order_id').size()
basket = basket.rename('basket_size')
orders = orders.merge(basket, on='order_id',
                      how='left', validate='m:1')
sns.histplot(data=orders, x='basket_size',
             discrete=True)
print(orders.shape, orders.basket_size.median())
```

```
(30229, 8) 2.0
```

In plain words

Read the order lines and count the lines on each order. That count is the basket size.

merge adds basket_size to each order. It now has one more variable.

sns.histplot draws the distribution of basket size, one bar per size.

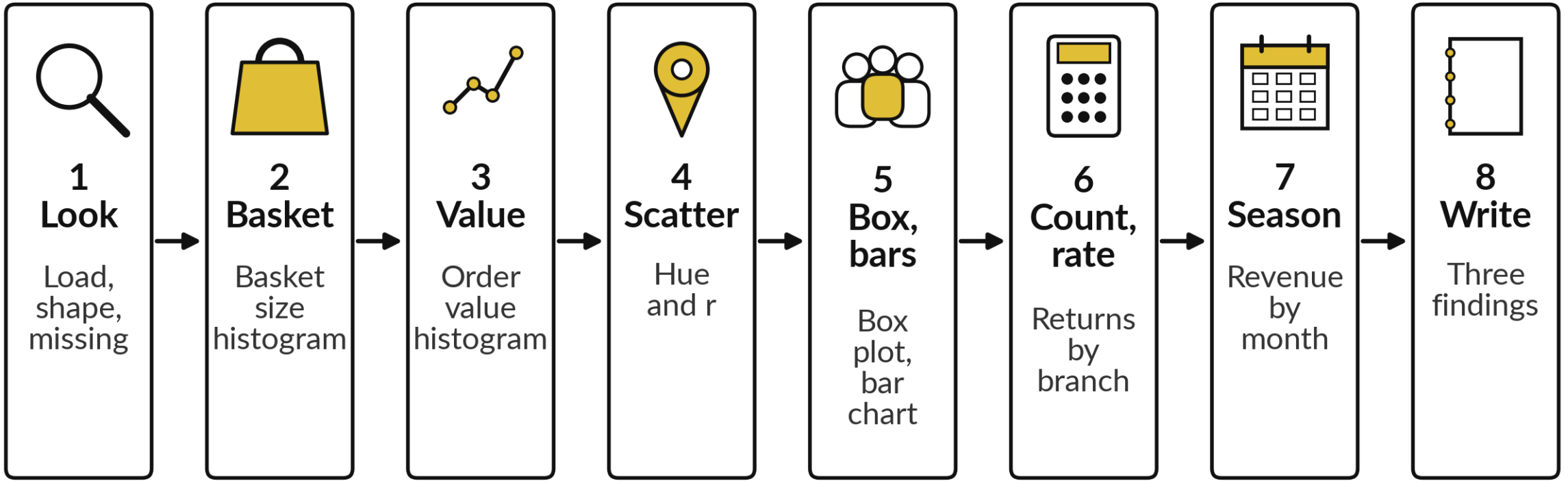
The median basket is 2 lines. The mean is 3.24, because a few baskets are large.



Part 2: the Python lab

Close the slides. Open the notebook.

The lab has eight steps, and you keep a notebook with eight charts and three findings



Twenty automatic checks. In each step: predict, run, read, change.

Check your numbers before you move on

What you compute	You should get
Records in orders, and missing customer_id	30,229 and 9,171
Median order value	23,200 UGX
Median basket size, and its largest value	2 and 35
Correlation of basket size and order value	0.30
Median order value of Card orders	23,825 UGX
Mean order value of branch B04	49,618 UGX
Returns at B01 and rate at B07 (per 100 lines)	321 and 3.18
December revenue over the other months, 2024 and 2025	1.55 and 1.26

If your number differs, find the step that went wrong. Do not change a number to match.

Five likely mistakes in the lab and how to fix them

	What you see	What it means	What to do
✗	**ValueError: Could not interpret value `branch` for `x`**	The column name does not exist in the DataFrame.	Print orders.columns and copy the exact name, such as branch_id.
✗	**KeyError: basket_size**	The merge result was not saved back.	Write orders = orders.merge(basket, on="order_id").
✗	**A chart with one tall bar and nothing else**	A few huge orders stretch the axis.	Set the axis limit, and say how many records are hidden.
✗	**The histogram shows no gaps between sizes**	Basket size counts whole lines, so bins should be whole numbers.	Add discrete=True.
✗	**NameError: name sns is not defined**	The import cell was not run.	Run the first code cell again, then the cell that failed.

Exit check: three questions, then compare with the short answers

	Question	Short answer
	**Branch B01 has the most returns. Does that make it the branch with the worst returns problem?**	Not yet known. Divide by lines sold to get the rate: B01 has 1.67 per 100 lines, the lowest, and B07 has 3.18, the highest.
	**Which chart answers "do two numeric variables move together?", and which seaborn function draws it?**	A scatter plot, drawn by sns.scatterplot. The correlation r puts a number on it.
	**Write the three parts of a finding.**	A number, a comparison and a limit.

Save your notebook with its eight charts and three findings

What you keep from today

Keep your exploration notebook: eight charts of the Lakeside orders and returns, and three findings written in plain words, each with a number, a comparison and a limit.

Our question today

How do I explore an unfamiliar dataset in one sitting and find what is worth reporting?

Next session

Session 12: Inference. When two groups differ, how do we know chance alone does not explain it? We test the loyalty offer.

Take away

Your lab file

Notebook, 8 charts

Your writing

3 findings

Next session

Is it real?